

Overall Effects of Global Warming

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Abstract

Global warming has many complex consequences around the world in many aspects of Earth's global ecosystem. The most profound of which is on the ocean, which is warming at a profound rate, caused by water absorbing heat from Earth's rising surface temperature. The heat trapped due to greenhouse gas effects is spreading throughout the ocean, causing damage to ecosystems, as well as human society. The effects are not limited to rising ocean levels, as the ocean is becoming more acidic, causing damage to ocean ecosystems, as well as corals becoming bleached. Kelp forests and marine life are also suffering due to these changes as well. Ocean warming is also having a catastrophic effect on glaciers, causing them to melt and thermometric ocean currents, where ocean warming is causing them to shift.

Introduction

Rising ocean temperatures are one of the most significant consequences of global climate change. Because the ocean absorbs most of the excess heat trapped in Earth's atmosphere, even small increases in temperature can disrupt systems in the ocean.

As temperatures rise, the ocean becomes more acidic, altering the chemical balance essential for many marine organisms. Coral reef systems face increasing stress and bleaching events, while fisheries that support millions of people experience shifting species distributions and declining productivity. Marine plant life, including phytoplankton that form the base of the ocean food web, may change in abundance and location, affecting marine animal life at every trophic level. At a larger scale, warming influences the thermohaline conveyor belt, the global circulation system that regulates climate by distributing heat across the planet. Meanwhile, the melting of Antarctic ice not only contributes to sea level rise but may further disrupt ocean currents and marine ecosystems.

Together, these impacts demonstrate that rising ocean temperatures are not simply an environmental concern but an issue with ecological, economic, and climatic consequences. Understanding these varying effects is essential to grasp the full scope of how a warming ocean reshapes the planet.

Temperature

Before we can talk about the effects of the ocean's rising temperatures, it is more important to understand how the temperature of the ocean has been changing over time.

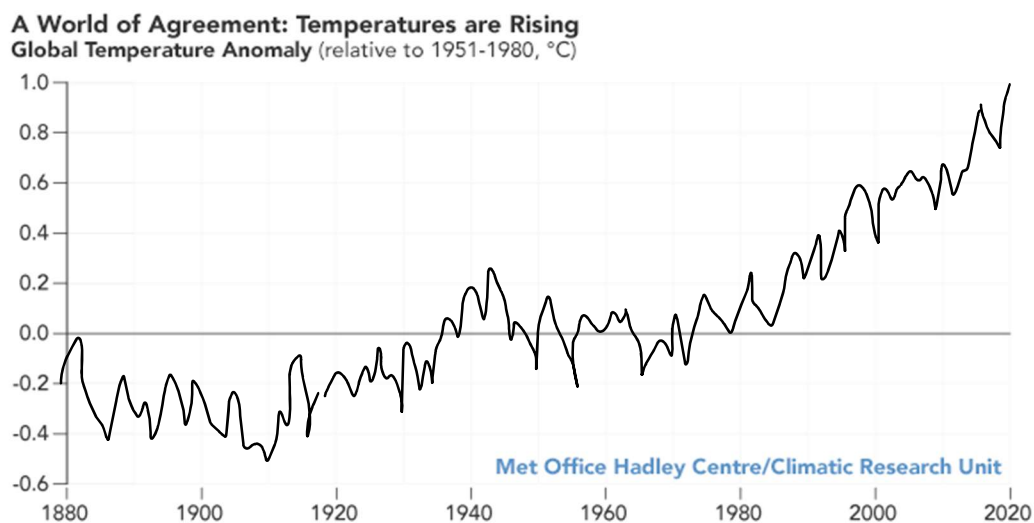


Figure 1: Global temperature anomalies from a 1950 average from 1880 to 2022. (Reference 1)

Global warming has been the main cause of rising ocean temperatures. Greenhouse gases produced naturally and expedited by industrial process and energy production get trapped in the atmosphere. These greenhouse gases allow energy from the sun to pass through the atmosphere, but when the heat tries to escape, the greenhouse gases absorb it, and release it back toward the earth, thus increasing the temperature of the earth. The ocean has absorbed 90% of the excess heat because of planetary global warming, causing ocean temperatures to rise. Water is known as a heat sink, meaning it has a high energy capacity. (reference 1) While global average temperatures have increased by 1.1 degrees Celsius from 1880

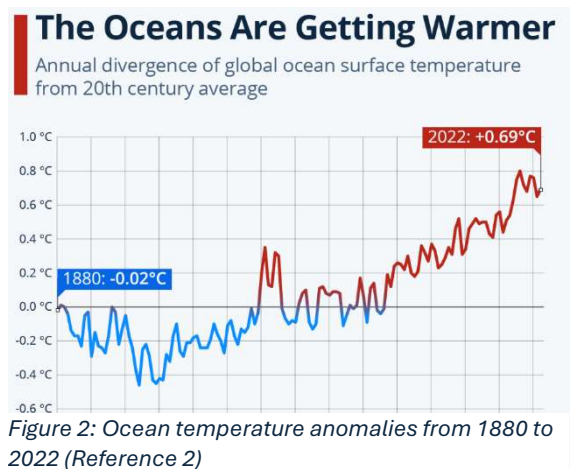


Figure 2: Ocean temperature anomalies from 1880 to 2022 (Reference 2)

to 2022, the average global ocean surface temperature has increased by 0.69 degrees Celsius. (reference 2) These changes in temperature may initially seem small, but on a global scale they have a large impact.

Acidity

When most people think about the effects of rising temperatures or global warming, ocean acidification is not the first thing that comes to mind. The common reason for this is because ocean acidification is not directly caused by rising ocean temperatures, instead it is caused by the same factors that cause increase ocean temperatures and global warming. The cause of global warming has more to do with rising concentrations in carbon dioxide. (CO_2)

Ocean acidification is a direct cause of a chemical reaction with atmospheric CO_2 and water. CO_2 occurs naturally in the environment, but its concentration in the environment has increased over time. One of the ways that CO_2 leaves the atmosphere is through the ocean. CO_2 can dissolve in water and when it does, a chemical reaction occurs. CO_2 reacts with water to create carbonic acid. This carbonic acid can disassociate in water, increasing the concentration of H^+ ions which decreases the pH of the ocean. (Reference 4) This increase in H^+ concentration also decreases the carbonate ion concentration, which is essential for marine life.

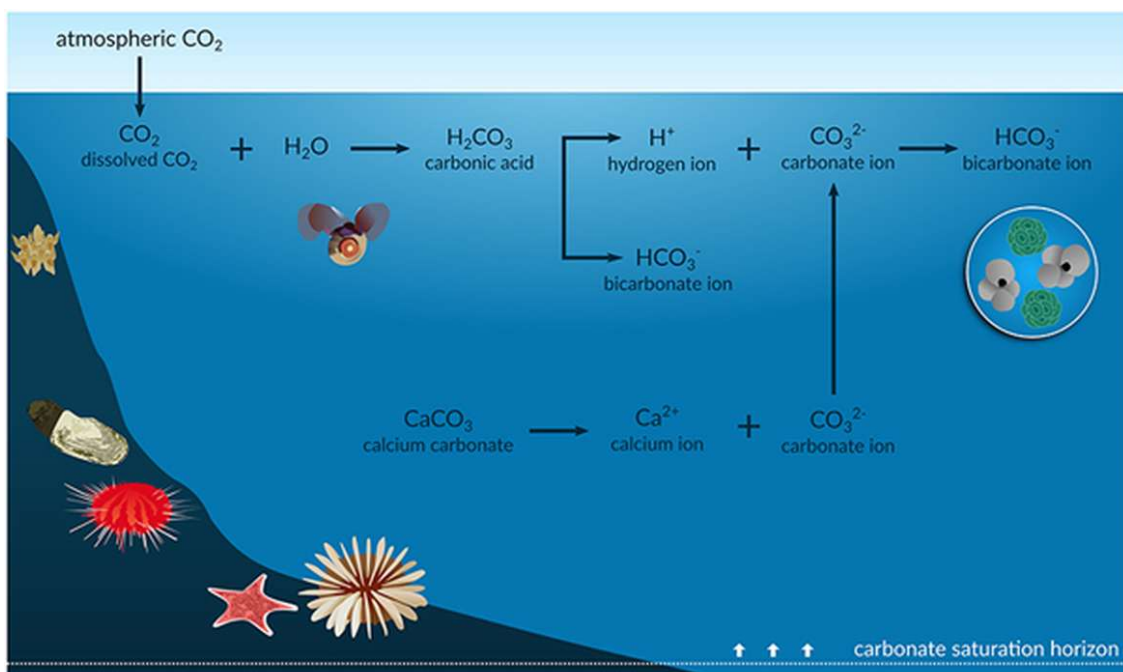
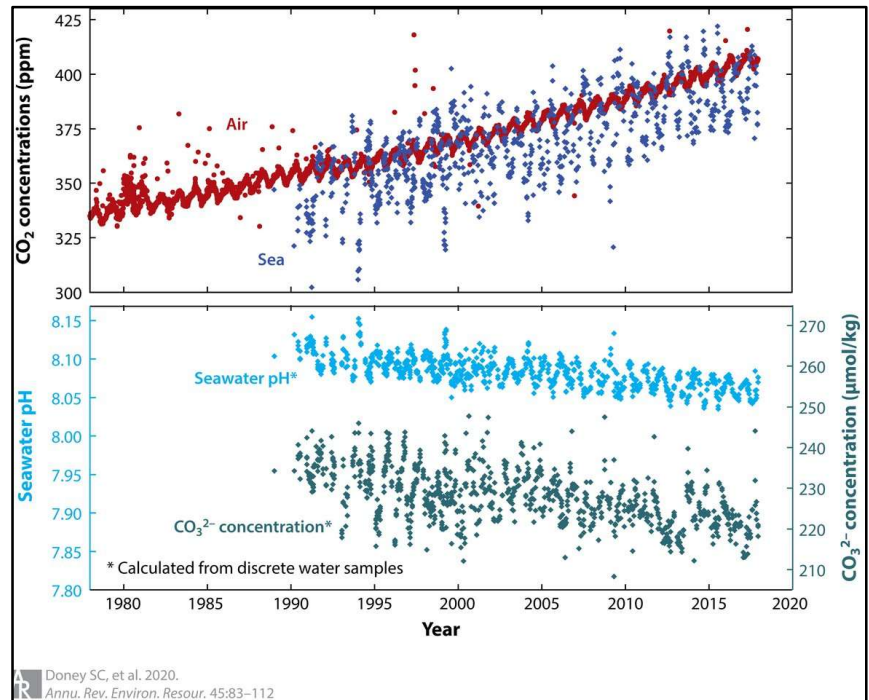


Figure 3: Diagram of Ocean acidification chemistry. (Reference 5)

Historically, the pH of the ocean has only increased. The historical jump in pH occurred corresponded with an increase in CO₂ Concentration. Because of this, pre-industrial pH levels are higher than they are in present day. In 1990 pH levels were 8.15 in present day those levels have decreased to 8.03. For reference, a decrease in pH of 0.1 refers to a 30% increase in H⁺ ion concentration. This correlates to the rate of CO₂ concentration increase from 1.8ppm/yr to 2.4 ppm/yr.



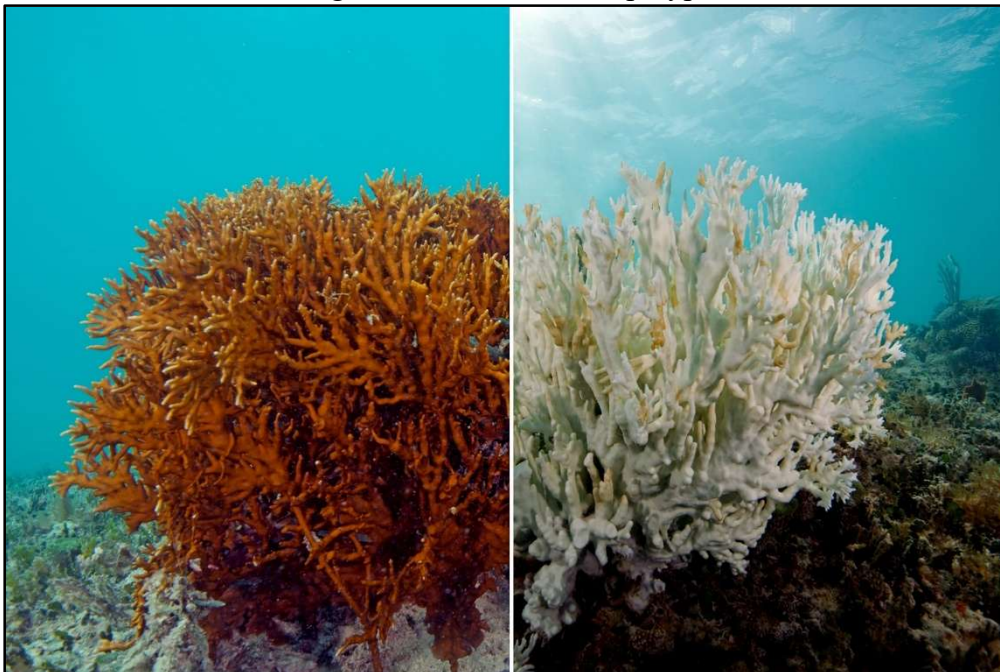
The acidification of the ocean has a lot of biological impacts as well. The reduction in carbonate ions in the ocean affects the lives of animals like shellfish and plankton. The kinds of animals use the carbonate ions in the ocean to create their shells in living structures, ocean acidification prevents this process, decreasing their survivability. This does not just affect shellfish in plankton, but the animals that consume them as well. The decrease in their population creates a cascade effect that can affect the ecosystem in a negative way. This decrease in carbonate levels also influences corals. However, corals face a much more direct threat from rising ocean temperatures.

Corals

To understand the effect that increasing ocean temperatures is having on corals, it is important to understand what corals actually are. There are two types of corals. Soft shell corals are thing like anemones and hard shell corals which build reefs. Increasing ocean temperatures mainly effect hard shell corals.

What looks like one large coral plant is a huge colony of small coral polyps. These polyps reproduce and join using the calcium carbonate they secrete as both a kind of cement and a shell to protect their soft bodies. These corals have a positive symbiotic relationship with a specific type of algae called zooxanthellae. The coral provides a safe space for the algae and in exchange the algae absorbs the CO₂ produced by the coral for photosynthesis and provides the corals with basic sugars and proteins that the coral needs to survive. Coral reefs are habitats for many fish as well, providing a safe and clean habitat for a wide variety of species including fish, eels, and rays. (Reference 7)

Coral bleaching is a term used to describe the visual effects that take place when the zooxanthellae inside the coral die or leave due to environmental stress. The algae are the component that give coral their color. So when they die the coral appears white as if it was bleached. A bleached coral is a good indicator that the polyps have died. The reason for this is



because if the algae inside the coral die, the coral polyps can no longer deceive nutrients or get rid of waste. When coral die, their calcium carbonate skeletons deteriorate and if the cause of the bleaching is widespread, it can lead to the destruction of large aquatic habitats.

Coral bleaching occurs when elevated temperatures disrupt the symbiotic relationship between corals and zooxanthellae. When stressed, corals expel these algae, losing both their primary energy source and pigmentation (Reference 9). Modern warming is occurring at a rate

unprecedented in the context of coral evolutionary history, limiting the ability of reefs to naturally adapt. Marine heatwaves have increased in frequency, duration, and intensity due to climate change. These prolonged temperature anomalies are now the dominant cause of mass coral bleaching events globally. For example, repeated bleaching events on the Great Barrier Reef were analyzed by Hughes et al. (2017) in *Nature*. Their study demonstrated that consecutive marine heatwaves in 2016 and 2017 caused severe bleaching across thousands of kilometers of reef. (Reference 10)

Historically, severe bleaching events were separated by decades, allowing ecosystems time to recover. However, the recurrence interval between bleaching events has shortened dramatically. This reduced recovery window results in cumulative stress, declining coral cover, and long-term ecosystem degradation. Coral reefs are among the most biodiverse ecosystems on Earth, supporting approximately 25% of marine species despite covering less than 1% of the ocean floor (Reference 11). When corals die, the structural complexity of reefs declines. This complexity provides habitat niches essential for fish, invertebrates, and microorganisms. Rapid climate change threatens to fundamentally reorganize reef ecosystems, potentially leading to widespread biodiversity loss.

Species that rely directly on live coral for food or shelter are particularly vulnerable, and localized extinctions may occur when coral mortality is high.

Fisheries

Marine fisheries play a critical role in global food security, employment, and economic stability. According to the Food and Agriculture Organization, approximately 3.3 billion people rely on fish as a significant source of animal protein, and hundreds of millions depend on fisheries and aquaculture for livelihoods. (reference 12) However, climate change is increasingly recognized as a major threat to marine fisheries productivity and stability.

As ocean temperatures increase, many marine species are shifting into deeper waters in search of suitable habitats. A global analysis by Pinsky et al. (2013), published in *Science*, found that marine species are moving in a way that corresponds to changing temperatures. (reference 13) These shifts can cause changes in regional fish availability, and increased competition over migrating species. The Intergovernmental Panel on Climate Change concludes with high

confidence that continued warming will cause substantial redistribution of marine fish stocks, especially toward higher latitudes. Ocean acidification directly affects calcifying organisms and indirectly affects fish through food web disruption. (reference 9)

Reduced carbonate ion availability impairs shell formation in mollusks and other calcifying organisms. Shellfish are among the most vulnerable marine organisms to acidification due to their dependence on calcium carbonate. Acidification has already caused larval oyster mortality events in parts of the U.S. Pacific Northwest, demonstrating real-world economic consequences.

Warmer water holds less dissolved oxygen, and increased stratification reduces oxygen replenishment in deeper layers. The IPCC reports widespread ocean deoxygenation linked to anthropogenic warming (reference 9). Hypoxic “dead zones” reduce available habitat for many fish species, forcing vertical compression of habitable space. When warming, acidification, and deoxygenation occur simultaneously, physiological stress intensifies (Reference 9). These compound stressors may reduce growth rates, increase disease susceptibility, and elevate mortality risk in both wild and farmed fish populations.

Climate-driven changes in fisheries distribution and productivity create economic instability and food security risks. It is projected that tropical developing countries may experience substantial reductions in fisheries quantity, while higher-latitude regions may see temporary increases. The IPCC (2022) further warns that shifting stocks may increase geopolitical tensions as fish cross national boundaries, complicating existing fisheries management agreements. (reference 9)

Marine Flora

Beyond its effects on coral, global warming also significantly affects other forms of marine plant life. Kelp forests are critical to ocean ecosystems around the world, providing food for sea urchins, abalone (a type of sea snail), and sea stars that feed directly on the kelp and the algae that grows on it. They provide protection from predators for small fish and invertebrates. For any creatures unlucky to be caught outside the kelp’s protection, a kelp forest is a rich buffet for sea otters, larger fish, and sea-diving birds. Unfortunately, due to rising sea temperatures worldwide, kelp forests are dying out in various ways. Rising temperatures have caused kelp

forests to shift worldwide. (Reference 10) This is because kelp cannot adapt to rising ocean temperatures in time; as new kelp plants populate, they vastly prefer warmer areas, and over time, the kelp forest moves. Any kelp that populates regions where the sea is warming will die out, leaving past sites of vast and mesmerizing kelp forests bare and uninhabited. This effect has been observed in the *Ecklonia Cava* species in Japan and the *Scytothalia Dorcarpa* in Australia. This shift has been observed mainly from coastlines into deeper waters, where kelp struggles to establish under the increased pressure (Reference 10). In total, any kelp that happens to populate past waters dies from heat, and any that populate new waters dies from pressure. This has led to the local extinction of some kelp species, such as the *L. Ochroleuca* and *S. Polyschides* in Spain. Furthermore, there is evidence that sea urchin grazing behavior increases under warming waters. Under normal conditions, sea urchins are a natural check for overgrowing kelp; however, under certain conditions, this isn't the case. In select species in tropical areas, kelp is taking over where coral is dying out, and with greater kelp abundance, sea urchins are proliferating at an exponential rate. While this is beneficial for kelp, the increased presence of both kelp and sea urchins decreases biodiversity in ocean ecosystems. This ultimately causes harm: the loss of several species creates an unstable ecosystem, increasing the risk that they will be lost in the coming decades. In conclusion, any significant changes to the kelp ecosystem affect ocean ecosystems and the global effects of rising ocean temperatures.

The increased carbon concentration within water has led to the deaths of numerous ocean species. All plants need carbon dioxide to survive, including ocean species such as the aforementioned kelp and coral; however, they need much less than their land counterparts. Carbon dioxide diffuses much more slowly into water and is less prevalent than in the general atmosphere. This has caused ocean plants to evolve to require less carbon dioxide than land plants, and with rising CO₂ levels, this poses an increased risk. Studies show that some ocean plants, specifically macroalgae (seaweed), are unable to adapt in time to rising CO₂ concentrations, whereas some species are (Reference 11). Some species of seagrass are showing an increased ability to absorb CO₂, and thus their metabolic growth rates have increased dramatically. These changes, as a whole, are predicted to affect ocean biodiversity drastically and, in turn, alter the food supply for ocean creatures. The sudden unavailability of some types of macroalgae will deprive some fish, like parrotfish, mullet, scrawled and filefish, of their food sources, as they feed only on local seagrass. Not only food, but shelter will be gone as well, as

some fish and crustaceans use seagrass as a place to raise their young. As local seagrass populations decline, several fish and other marine species will be left without their natural habitat and eventually die out.

Marine Fauna

Moving on to marine life, the effects are similar. As previously discussed, rising Earth temperatures have caused plant life to die out and biodiversity to decline. There is a similar effect is occurring among marine creatures, where climate change is significantly affecting their way of life. Research has examined what is happening to marine animals at the cellular and genetic levels due to climate change (Reference 13). The results were that not even the smallest expressions of life were safe from humanity's chokehold. Gene expression is a complex biological process, but one surprising fact is that the environment can influence it. The environment around a creature can directly affect how specific genes are expressed; for example, a tallness gene may not be expressed in someone who is malnourished. This can also happen with temperature. What researchers found was that increased water temperatures have caused populations of small ocean organisms (shrimp, krill, plankton, etc.) to die out due to muscular deficiency. These highly mobile organisms need to be on the move constantly in search of food. Without this ability, not only do they starve, but they also cannot escape oncoming predators. It also affected reproduction in some populations, leading to decreased brood sizes. For those lucky enough to be born, the pre-adult stages lasted longer, making them more susceptible to predators they could not outrun. All of this was caused by genes not activating correctly due to rising temperatures, as the proteins responsible for carrying out genetic functions were unfamiliar with the sudden temperature increase in the surrounding water and became dormant. The researchers hypothesized that either the proteins mistook these environmental factors for immunosuppression or were encoded only at specific depths, where the water is usually cooler. Overall, the research clearly shows that small marine creatures are unable to cope with the ongoing impacts of climate change.

Furthermore, the effects of climate change on marine life are being felt worldwide. It is easy to dismiss the potential impact of climate change as local or too small to study, but a 2013 study proved beyond a doubt that these effects are global. What their research showed was that

81-83% of ocean species responded as predicted to the impact of climate change. In Figure 7 (reference 12), the circles represent population centers of marine life, and as shown, most centers are almost entirely as predicted. Around the world, marine animal species are being affected. To

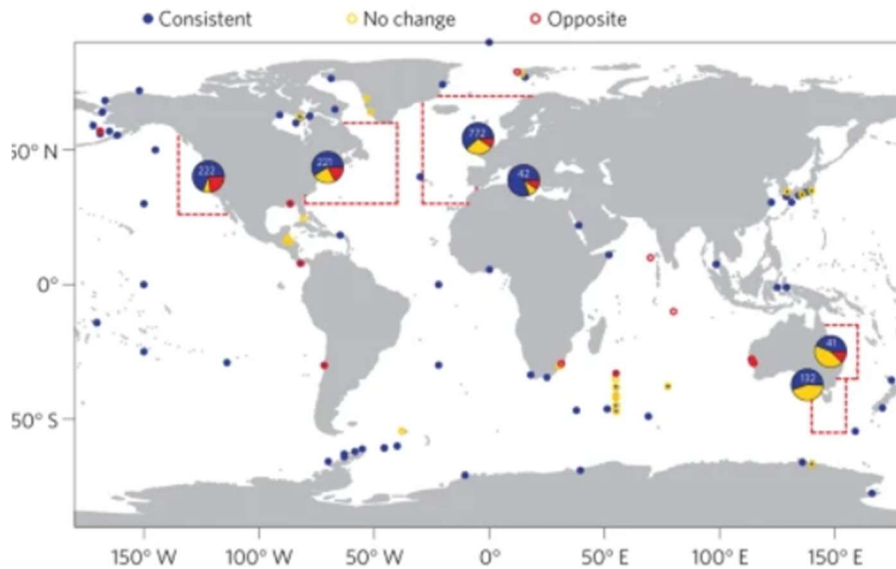


Figure 7: Chart detailing centers of effect due to climate change (Reference 12)

avoid statistical bias in their research, scientists studied previous findings dating back decades to determine if their findings were significant. Their study fell right in line with what climate scientists have long predicted: that

populations upon

populations of ocean creatures have been dying out. This affected their findings, potentially introducing publication bias: conclusions of a group of researchers may not be published because they're not significant enough. To combat this, they highlight insignificant data points, as shown in the graph above. As observed in Australia in Figure 7, there is little change in ocean populations there. This is most likely because ocean creatures in the Australian region are better acclimated to warmer climates and, as such, can tolerate higher temperatures more easily.

Moving past organisms, climate change has had a highly publicized effect on the polar ice caps. As the Earth warms up, more and more of the polar ice caps melt every year. This causes higher ocean levels and habitat loss for Arctic creatures. Furthermore, it spells doom for Arctic indigenous tribes in northern Canada. The tribes in these northern regions rely on these melting glaciers for daily life. Cultural and religious practices are directly tied to the glaciers, which are considered sacred and essential to theological practices. They use them for worship and to feel connected to the land they inhabit. Spiritual practices are vital to their religion; without them, they are forced to relocate to continue their belief systems. With the glaciers gone, the cultural beliefs they have practiced for thousands of years will die out, becoming another casualty of climate change. (Reference 14) To people who practice these beliefs, the melting

glaciers represent a complete breakdown of their beliefs, and they are unable to comprehend the actual reason for their disappearance as the fault of a giant megacorporation or industrial nation thousands of miles away. They also rely on them for food, as ice fishing is an essential source of sustenance in many of these communities, and for migration, where nomadic communities depend on glaciers for their way of life. Without the glaciers, these migrating communities have no natural bridges between the lands they inhabit and thus cannot maintain the lifestyles they have practiced for thousands of years.

Antarctic Ice Melting

Since 1880, global sea levels have risen by 8-9 inches (Reference 20). Over 30 years, from 1993 to 2023, 3.99 inches of water were added to the ocean, indicating an average sea level rise of 0.133 inches per year, more than double the rate throughout most of the 20th century (Reference 20).

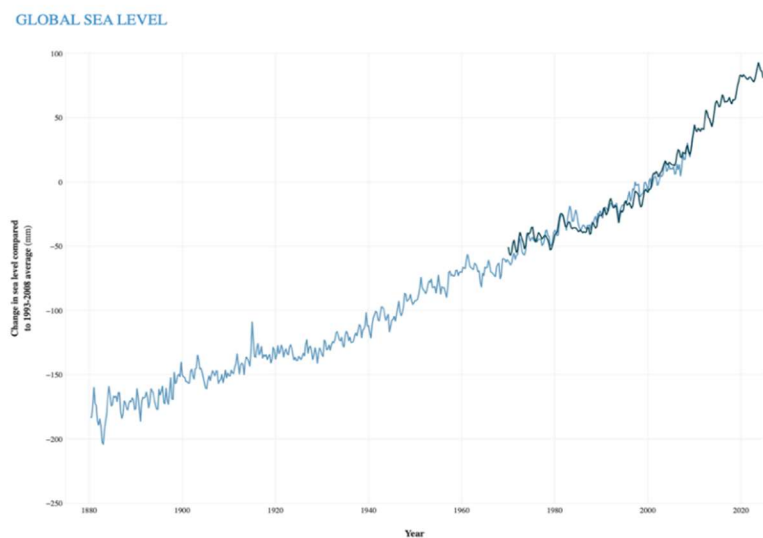


Figure 9: Change in sea level based on year 2000 (Reference 20)

From the 1970s to the 2010s, melting of ice and the heat-induced expansion of water had contributed roughly equally to global sea level rise (Reference 20). However, Greenland Ice Sheet's loss increased sevenfold from 34 billion tons per year between 1992-2001 to 247 billion tons per year between 2012 and 2016; Antarctic ice loss nearly quadrupled from 51 billion tons per year between 1992 and 2001 to 199 billion tons per year from 2012-2016 (Reference 20). These data underpinned the situation in which ice-melting had doubled the heat expansion from 2005 to 2013 and is expected to increase exponentially in the future.

According to NASA, Antarctica added approximately 0.41mm per year from 2002 to 2020, which is about 12% of all increases. Based on terrain, Antarctica is divided into 2 main

regions: the West Antarctic Ice Sheet (WAIS), which is mostly grounded below sea level, and the East Antarctic Ice Sheet (EAIS), which is mostly grounded above sea level.

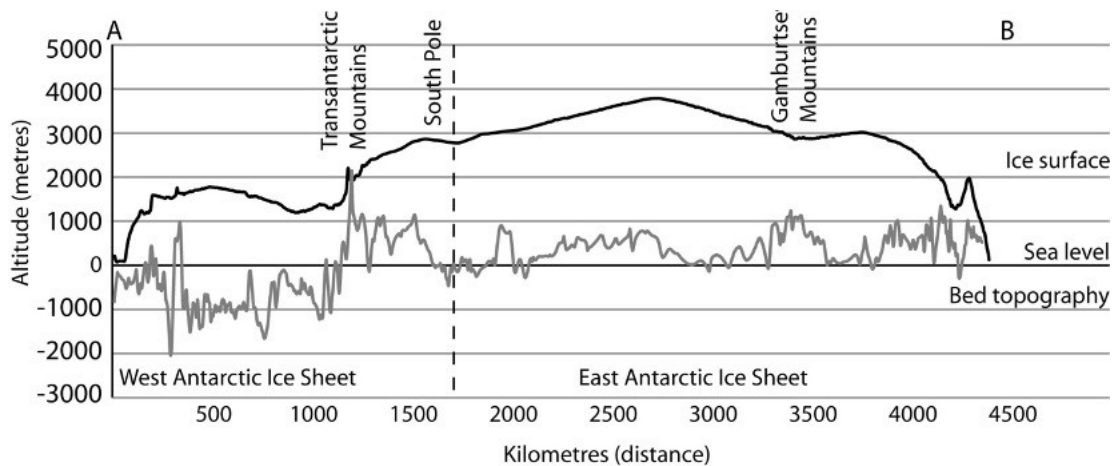


Figure 10: Altitude of Antarctica from west to east (Reference 21)

Altitude of Antarctica from west to east

Most of the ice melting in Antarctica comes from the WAIS, especially through the Pine Island Glacier and Thwaites ice stream into Pine Island Bay (Reference 23). It loses mass through evaporation, melting at its surface (top, edges, and bottom), and calving of ice shelves that form icebergs, which increases the surface area in contact with warmer water (Reference 21). This imbalance is also attributed to warm ocean currents—modified Circumpolar Deep Water (CDW), a saltier and warmer water mass in the Pacific and Indian Ocean, causing grounding-line (line where the ice sheet lifts off and begins floating) retreat and marginal ice thinning, which further leads to reduced buttressing, increasing ice discharge (Reference 20).

Observation of shelf-water temperature trends is extremely sparse around EAIS, with few multi-decadal measurements available (such as in the Ross Sea) (Reference 22). Evidence for long-term shelf-water warming is therefore limited. However, warm waters have been detected near several outlet glaciers, coinciding with high basal melt beneath ice shelves (Reference 22).

The Marine Ice Sheet Instability hypothesis proposes that the grounding line in WAIS can retreat rapidly due to its downward slope towards the inland. This is because, as warm ocean water melts the ice at the grounding line, thicker ice will sit in deeper water, exposing more ice

to water and accelerating melting. This process can drive further recession of the grounding line, creating a positive feedback loop. Estimates suggest a 5% probability that the WAIS will contribute 10 mm of sea-level rise within 200 years (2400% of the current rate) (Reference 23).

Atlantic Thermohaline Conveyor Belt

The Atlantic thermohaline conveyor belt, or more specifically, the Atlantic meridional overturning circulation (AMOC), is a vast oceanic loop that carries warm water from the Atlantic equatorial region northward to the uppermost Atlantic, toward Iceland and Greenland, where it cools and descends before returning southward (Reference 25). Ocean currents are mainly driven by salinity and temperature: high salinity and low temperature make water sink.

The Atlantic is the only ocean that produces deep water in the Northern Hemisphere. As the Gulf Stream transports warm water northward from the Gulf of Mexico under wind forcing, it reaches the Norwegian and Labrador Seas, where surface waters cool through heat loss to the atmosphere (Gordon 46). Cooling increases seawater density, causing it to sink and form deep water masses called North Atlantic Deep Water (NADW). This convection process initiates the vertical thermohaline circulation. The sunken water then moves south at a depth of 2–3 km until joining the Antarctic Circumpolar Current (Reference 25).

Unlike the Pacific and Indian oceans, which move heat from the tropics to high-latitude regions in both hemispheres, the Atlantic transports all the warm water northward at all latitudes, even from the southern hemisphere (Reference 24). The figure below, by NASA, illustrates the oceanic current dynamics on Earth. Consequently, the northern Atlantic regions, especially Western Europe, receive sufficient warmth and experience milder weather. By comparison, the North Atlantic is about 4°C warmer than the Pacific at comparable latitudes (Gordon 46). Hence, the functioning of the AMOC is instrumental in regulating the climate.

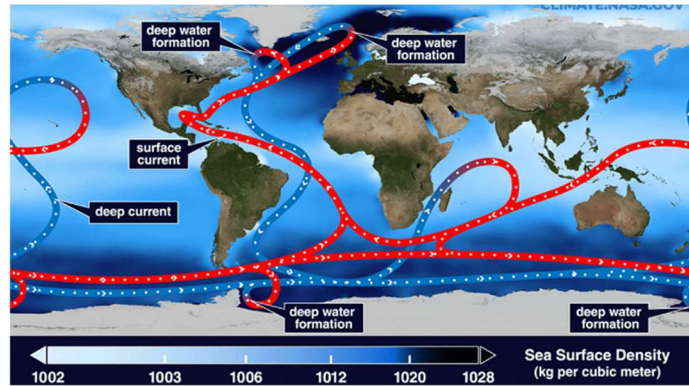


Figure 11: Global thermohaline conveyor belt (Reference 24)

However, ice melting driven by global warming may be capable of triggering AMOC collapse. According to NASA, Greenland shed approximately 264 gigatons of ice per year between 2002 and 2025. This fresher and purer meltwater overtops and displaces saltier water from the AMOC. Even though the meltwater is cooled, its low salinity causes it to float on the surface rather than joining the AMOC's descending branch, impeding the flow of the warm oceanic current cycle and thus reducing the convection rate (Reference 27). If the situation were to be exacerbated and the North Atlantic Conveyor were to shut down in the future, the stoppage of 15 to 20 million m³/s warm water (Reference 27) flowing to the UK and northern Europe would roughly lead to a 5°C mean annual temperature drop in these regions (Reference 26).

The possibility of such ice-age conditions, similar to the Younger Dryas, returning to Europe is of great societal relevance and needs to be addressed. This is because the concept of an impending shutdown of the SMOC is largely based on measurements of increasing freshness of the deep North Atlantic Ocean since the 1950s, and on up to 20% reductions in flow in some northern conveyor belt areas (Reference 26). Estimation suggested that the possibility of a further weakening rate is at 2-4% per year (Reference 26).

Nevertheless, accurate and reliable measurements of the strength of the northern conveyor are still significantly limited by scientists' technological capacity due to its massive size and temporal nature. Additionally, data acquired over the last several decades is not sufficient to account for the unpredictable short-term factors. For example, after a weak phase in the late 1960s, convection in the Northern Conveyor, especially in the Scottish region, increased again in the 1990s, resulting in unexpected devastation of local marine food chains and the loss of breeding ability for hundreds of thousands of seabirds (Reference 26).

Multiple models used by scientists have consistently indicated a significant weakening of the conveyor belt's flow throughout the 21st century due to global warming. For example, the model by the Canadian Center for Climate Model and Analysis (CCC) estimates a significant decline of up to 50% by the end of the 21st century. However, no model suggests the likelihood of a total collapse in the future (Reference 26).

Conclusion

Global warming, as a result of industrialization, has many implications in many aspects of the world. In this paper, we discussed the effects of global warming on plant life, corals, fisheries, marine animals, the Atlantic thermohaline conveyor belt, and Antarctic ice melting. The acidification of seawater impaired the formation of shells by marine organisms such as corals. Warmer ocean temperatures deoxygenate seawater, affecting the dynamics and growth rates of fisheries. Besides that, the melting of glaciers and ice sheets in Antarctica and Greenland added substantial freshwater to the ocean, not only raising sea levels but also impeding the Atlantic thermohaline conveyor belt. Even though the effects of global warming have not substantially affected daily life, this remains a long-term, insidious global problem that spans every corner of the world and will continue to worsen until its impact becomes significant enough to raise public awareness and enable effective change.

Recommendations

Solving global problems is not an individual country's challenge; it can only be tackled effectively if all countries work together. Hence, the world must focus on reducing carbon footprints. When it comes to policy, the government should provide tax incentives to businesses and households to invest in renewable energy infrastructure to replace fossil fuel plants. On an individual level, switching off the light and air conditioner when not in use can help reduce overall electricity use. Replacing worn-out home electrical appliances with more efficient models can also significantly reduce electricity consumption.

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